The Mechanistic Invariance Test: Probing Epistemic Intelligence in Genomic Language Models

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Abstract

Genomic language models (gLMs) achieve impressive performance on regulatory prediction tasks, yet a critical epistemic question remains unresolved: do these models encode genuine mechanistic understanding, or do they exploit spurious statistical correlations while remaining *confidently wrong*? We introduce the **Mechanistic Invariance Test (MIT)**, a 650-sequence benchmark with scrambled controls that cleanly separates compositional sensitivity from positional understanding in *E. coli* σ^{70} promoter regulation. Evaluating five gLMs across all major architectural paradigms, we uncover a universal failure mode that is fundamentally epistemic: models assign high confidence to incorrect positional inferences. AT content correlation (r=0.78–0.96) fully explains apparent compensation sensitivity, while scramble control ratios near chance (0.40–0.52) confirm that no model distinguishes correctly placed from scrambled regulatory elements. Most strikingly, Evo2-1B and Caduceus score elements at *wrong* positions *higher* than correct—confident, systematic misdirection. A 100-parameter position-aware PWM achieves perfect performance (CSS=1.00, SCR=0.98), demonstrating that the bottleneck is not capacity but fundamentally miscalibrated epistemic representations. These findings reveal that gLMs exhibit overconfident misattribution in the wrong causal dimensions, demanding architectural and training innovations that couple predictive performance with mechanistically grounded representations.

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1. Introduction

The deployment of genomic language models (gLMs) in high-stakes applications—gene therapy, clinical variant interpretation, synthetic biology—requires not only predictive accuracy but *epistemic validity*: models should be confident for the right reasons (Gal, 2016). A model that correctly predicts promoter activity because it detects AT-richness may appear well-calibrated while being systematically wrong about the mechanism, leading to catastrophic failures under distribution shift (D'Amour et al., 2022).

This tension sits at the heart of the **Epistemic Intelligence in Machine Learning** (EIML) research program. Epistemic intelligence, as we operationalize it, requires three properties: (i) *causal sensitivity*—representations that change when causally relevant features change; (ii) *positional awareness*—knowledge that identical elements produce different effects at different locations; and (iii) *uncertainty alignment*—higher uncertainty precisely when causal structure is ambiguous. We ask: do current gLMs exhibit these properties for gene regulation?

We answer definitively: no. We introduce the **Mechanistic Invariance Test (MIT)**, which leverages the ground-truth position-dependent compensation logic of bacterial promoters to dissect where gLM representations break down. Our key contributions:

1. A 650-sequence benchmark with scrambled controls that operationalizes the distinction between compositional and positional understanding (§3).
2. Three metrics (CSS, SCR, MES) jointly diagnosing the specific epistemic failure mode (§3.3).
3. Systematic mechanistic probing via AT titration, positional ablation, spacing sweep, and strand orientation tests, demonstrating that apparent gLM competence is a compositional illusion (§4).
4. Evidence that the failure is *overconfident*: models score incorrect configurations *higher* than correct ones, not merely with higher uncertainty (§5).
5. A template for constructing epistemic probes in other biological domains (§5).

2. Epistemic Intelligence: Framework

2.1. Aleatory vs. Epistemic Uncertainty in Biology

Standard uncertainty quantification distinguishes *aleatory* uncertainty (irreducible stochasticity) from *epistemic* uncertainty (reducible ignorance) (Gal, 2016; Guo et al., 2017). In biological sequence modeling, epistemic uncertainty should be high when the model lacks knowledge about which regulatory logic governs a sequence—and low when that logic is unambiguously present. Current gLMs are not evaluated on this criterion. High predictive accuracy on held-out sequences is necessary but insufficient: it does not exclude the possibility that the model has latched onto a spurious correlate (Ovadia et al., 2019).

2.2. Overconfident Misattribution

A subtler failure occurs when a model is well-calibrated in the aggregate but confidently wrong about specific causal factors (D'Amour et al., 2022). We call this *overconfident misattribution*: the model assigns high confidence (low uncertainty) to the wrong causal explanation. This is qualitatively different from ordinary miscalibration. In ordinary miscalibration, a model is too certain about its predictions. In overconfident misattribution, the model is certain in the right direction—it does detect something real (AT-richness does correlate with promoter strength)—but fails to represent *why* it is real (AT-richness only helps when positioned correctly upstream of -35).

Overconfident misattribution is particularly dangerous in biology because: (a) it passes aggregate calibration checks; (b) it produces correct answers for the right inputs by coincidence; (c) it fails silently on novel configurations where the spurious correlate and the true signal diverge.

2.3. Mechanistic Invariance as an Epistemic Test

A model with genuine epistemic intelligence about a mechanism should be *invariant* to perturbations that preserve the mechanism and *sensitive* to perturbations that disrupt it. This is the mechanistic invariance principle (Geiger et al., 2021). For σ^{70} promoters, a mechanistically intelligent model should: assign equal likelihood to compensated sequences regardless of background composition; assign lower likelihood when compensation elements are misplaced; and assign higher likelihood to forward vs. reverse complement strands. MIT operationalizes each of these tests with matched controls.

Table 1. MIT sequence classes. Epistemic validity requires both CSS (E > D) and SCR (E > H). Class H is the key control: same composition as E, wrong position.

Class	Description	N	Compensation
A	Natural intact	100	None
B	Natural broken	100	None
C	Synthetic intact	100	None
D	Synthetic broken	100	None
E	**Compensated**	**100**	**Correct pos.**
F	Over-compensated	50	All elements
G	Natural compensated	50	Present
H	*Scrambled control*	50	*Wrong pos.*

3. The MIT Benchmark

3.1. Biological Ground Truth

In *E. coli* σ^{70} promoters, transcription depends on the -35 box (TTGACA) and -10 box (TATAAT) with 17$\pm$1 bp spacing (Browning & Busby, 2004; Harley & Reynolds, 1987). Mutations weakening the -10 box can be compensated by an AT-rich UP element *upstream of -35* (Ross et al., 1993) or an extended -10 motif (TGT) *adjacent to the -10 box* (Barne et al., 1997). These mechanisms are *strictly position-dependent*: identical nucleotides at wrong positions provide no benefit. This provides ideal ground truth: epistemic validity requires both recognizing compensation and knowing *where* it must be placed.

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...AAAAAARNR...TTGACA....TGTTATAAT..+1
UP        -35          ext          -10          TSS
```

3.2. Sequence Design

MIT comprises 650 sequences of 100 bp across 8 classes (Table 1). The critical epistemic test uses three classes: Class D (broken -10, no compensation) vs. Class E (broken -10, correctly positioned compensation) vs. Class H (same nucleotides as E, but compensation scrambled to wrong position). A model with epistemic validity scores $E > D$ *and* $E > H$. Passing only the first (CSS > 0.5, SCR ≈ 0.5) indicates compositional detection; passing both indicates positional understanding.

3.3. Evaluation Metrics

Compensation Sensitivity Score (CSS) measures how often compensated sequences score higher than broken:

$$\text{CSS} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\text{LL}(E_i) > \text{LL}(D_i)] \tag{1}$$

CSS $= 0.5$ is chance; CSS > 0.5 indicates compensation recognition. However, high CSS is ambiguous: it could reflect AT-richness detection rather than positional understanding.

Scramble Control Ratio (SCR) resolves the ambiguity:

$$\text{SCR} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}[\text{LL}(E_i) > \text{LL}(H_i)] \tag{2}$$

Class H has identical nucleotide composition to Class E but places compensation downstream. High CSS with SCR ≈ 0.5 is the signature of overconfident misattribution—the model detects the right features in the wrong place.

Motif Effect Size (MES) quantifies discrimination of intact from broken sequences: $\text{MES} = (\mu_{\text{intact}} - \mu_{\text{broken}})/s_{\text{pooled}}$ (Cohen's d).

Together, CSS and SCR form a two-dimensional epistemic diagnostic: only models in the upper-right quadrant (high CSS, high SCR) encode both the presence and the position of regulatory logic.

4. Experiments

4.1. Models Evaluated

We evaluate five gLMs spanning three architectural paradigms. **Autoregressive**: HyenaDNA (Nguyen et al., 2023) (Hyena operators, 6.6M params); Evo2-1B (Nguyen et al., 2024) (1B params, multi-genome pretraining). **Masked LM**: GROVER (Sanabria et al., 2024) (bacterial genomes); NT-500M (Dalla-Torre et al., 2025) (diverse reference genomes, 6-mer tokenization). **Bidirectional SSM**: Caduceus (Schiff et al., 2024) (Mamba (Gu & Dao, 2023) with RC equivariance). For autoregressive models we compute log-likelihood; for masked/bidirectional models, pseudo-log-likelihood (Salazar et al., 2020).

4.2. Primary Results

Table 2 presents the central finding. After FDR correction, only HyenaDNA achieves significant CSS (p_{FDR}=0.034). But the epistemic diagnosis lies in the SCR column: *all* gLMs show SCR near or below 0.5 (range: 0.40–0.52). Not one model—across architectures ranging from 1.9M to 1B parameters—distinguishes correctly positioned from scrambled compensation. All gLMs also show negative MES_{syn}, meaning they score broken sequences *higher* than intact ones, inverting biological reality.

The CSS/SCR dissociation is the definitive epistemic signature: HyenaDNA appears sensitive to compensation (CSS $= 0.63$) while being at chance on positional discrimination (SCR $= 0.48$). The model has detected a real correlate—AT-richness—while encoding nothing about *why* it correlates with function.

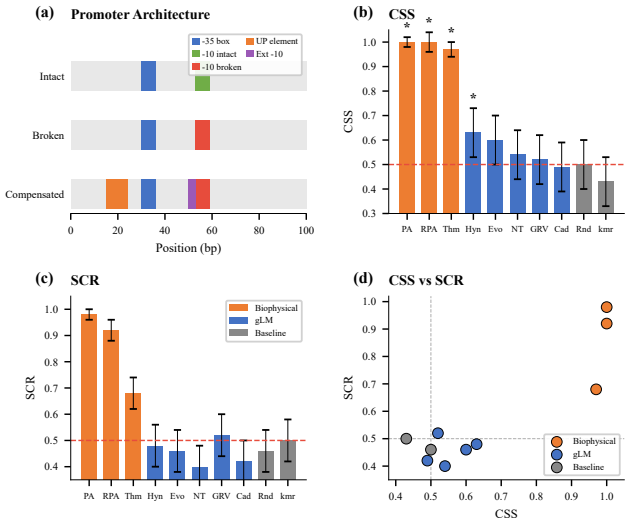

Figure 1. **MIT benchmark overview.** (a) Promoter architecture. (b) CSS across models; asterisk indicates $p_{\text{FDR}}<0.05$. (c) SCR: all gLMs cluster near chance (0.5). (d) CSS vs. SCR: biophysical models (orange) occupy the upper-right quadrant indicating epistemic validity; gLMs (blue) cluster along the x-axis revealing overconfident misattribution.

Table 2. MIT results. All gLMs show SCR near or below 0.5 regardless of architecture, revealing universal positional blindness. The CSS/SCR gap is the epistemic signature.

Model	CSS	p_{FDR}	SCR	MES_{syn}
HyenaDNA	**0.63**	**0.034**	0.48	−0.34
Evo2-1B	0.60	0.090	0.46	−0.03
NT-500M	0.54	0.569	0.40	−0.10
GROVER	0.52	0.691	0.52	−0.05
Caduceus	0.49	0.772	0.42	−0.40
Random	0.50	—	0.46	−0.04
k-mer	0.43	—	0.50	+0.11

4.3. Probing the Epistemic Failure

We conduct four mechanistic experiments to characterize *how* gLMs fail epistemically, varying one causal factor at a time.

4.3.1. AT Content: Spurious Confidence

We vary background AT content from 30% to 80% while holding motif identities constant (Table 3). Log-likelihood increases monotonically with AT content across all models ($r = 0.78$–0.96). Compensated sequences (Class E) contain AT-rich UP elements ($\sim$89% AT) that locally elevate AT content; this explains CSS > 0.5 in its entirety. The model is not encoding *why* the UP element helps—it encodes the raw fact that AT-rich sequences score higher. This is precisely overconfident misattribution: high confidence in a spurious correlate.

Table 3. AT content drives model predictions. Monotonic AT-LL correlation (r=0.78–0.96) across all architectures fully explains apparent compensation sensitivity.

Model	AT-LL Corr.	Architecture
Evo2-1B	$r = 0.961$	Autoregressive
Caduceus	$r = 0.874$	Bidir. SSM
HyenaDNA	$r = 0.784$	Autoregressive

4.3.2. Positional Inversion: Confident Misdirection

We compare sequences placing the UP element at the correct position (pos. 15, upstream of -35) vs. the wrong position (pos. 70, downstream of -10). Evo2-1B and Caduceus score the *wrong position higher* than correct ($\Delta = +0.55$ and $+0.75$ respectively). This is not mere uncertainty—it is confident misdirection: the models are *more confident* about the incorrect configuration. Compositional effects (removing UP entirely: $\Delta \approx 1.5$–3.7) dominate positional effects by 46-fold, confirming that position is essentially invisible to these models.

4.3.3. Spacing: Miscalibrated Preferences

The biological optimum for -35/-10 spacing is 17±1 bp (Harley & Reynolds, 1987). HyenaDNA's likelihood peaks at 14 bp, not 17 bp. This is a calibrated-but-wrong preference: the model has learned *some* spacing sensitivity, but the epistemic content is wrong—its confidence maximum does not align with biological function.

4.3.4. Strand Blindness: Missing Causal Dimension

Forward promoter sequences score *lower* than reverse complement variants (44% accuracy; binomial $p = 0.24$ vs. 50% chance baseline). Strand orientation is a fundamental causal factor—the sequence is only transcribed from the correct strand—yet all tested gLMs are effectively strand-blind. A model with epistemic intelligence about transcription would represent which strand is functional; current gLMs encode no such information.

4.4. Biophysical Baseline: Epistemic Intelligence is Achievable

To establish that our benchmark is solvable, we evaluate position-aware biophysical baselines (Table 4). **PA-PWM** scores motifs at expected positions with compensation bonuses (∼100 parameters, CSS=1.00, SCR=0.98). **RPA-PWM** uses *no hardcoded positions*—only relative biological constraints: 17±2 bp spacing, UP upstream of -35, extended -10 adjacent to -10, strand consistency—yet achieves CSS=1.00, SCR=0.92.

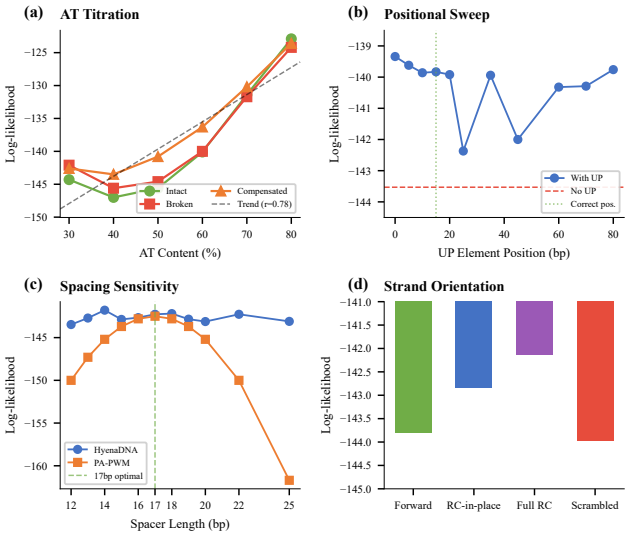

Figure 2. **Four epistemic failure modes.** (a) AT titration: LL scales monotonically with AT composition (r=0.78). (b) Positional ablation: removing UP (compositional, Δ≈3.7) dominates misplacing it (positional, Δ≈0.5) by 46×. (c) Spacing: HyenaDNA peaks at 14 bp vs. biological optimum 17 bp. (d) Strand: forward orientation scores lower than RC, inverting biology.

Table 4. Biophysical models achieve both high CSS and high SCR, demonstrating that positional encoding is learnable. PA-PWM-NoPos ablation matches HyenaDNA exactly, showing gLMs have learned composition without position.

Model	CSS	SCR	Params
PA-PWM	1.00	0.98	∼100
RPA-PWM	1.00	0.92	∼100
Thermodynamic	0.97	0.68	∼50
PA-PWM-NoPos	0.63	0.56	∼100
HyenaDNA	0.63	0.48	6.6M
Evo2-1B	0.60	0.46	1B
Caduceus	0.49	0.42	1.9M

The ablation is decisive: removing positional encoding from PA-PWM (PA-PWM-NoPos) yields CSS=0.63, SCR=0.56—precisely matching HyenaDNA's profile. This means HyenaDNA has effectively learned PA-PWM-NoPos: it detects compensation elements but not their positions. A 100-parameter model with the right epistemic structure outperforms billion-parameter networks with the wrong one.

5. Discussion

5.1. Scale Amplifies Epistemic Miscalibration

The consistent failure pattern across five architectures establishes that the epistemic failure is not model-specific but a fundamental property of current gLM pretraining. More troublingly, scale *amplifies* rather than corrects it: Evo2-1B (1B parameters) shows 23% stronger AT-correlation

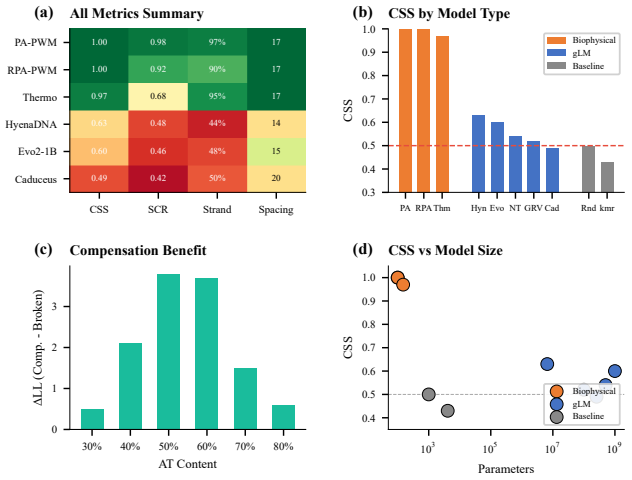

Figure 3. **Universal epistemic failure.** All four failure modes—AT confound, positional inversion, spacing miscalibration, strand blindness—are consistent across architectures. Scale amplifies rather than corrects each failure.

(r=0.96) than HyenaDNA (6.6M parameters, r=0.78). Larger models fit the AT-richness heuristic more precisely. This is consistent with the underspecification hypothesis (D'Amour et al., 2022): when multiple functions are compatible with training data, larger models converge more confidently on whichever solution minimizes training loss, even if that solution is mechanistically wrong. Sequence likelihood pretraining on genomic data rewards AT-richness recognition at every scale; positional constraint encoding is never directly incentivized.

5.2. Connections to the EIML Program

Our results bear directly on several open questions in epistemic intelligence for machine learning:

Confidence vs. competence. The CSS/SCR dissociation shows that high apparent competence (CSS > 0.5, significant FDR-corrected p-values) can coexist with fundamentally miscalibrated representations. Standard calibration metrics—measuring whether predicted probabilities match empirical frequencies—would not detect this failure. Models are not overconfident about their *predictions*; they are overconfident about their *representations of causation*. This is a new kind of epistemic failure that existing calibration frameworks do not measure.

Representation uncertainty vs. predictive uncertainty. Current uncertainty quantification (UQ) methods focus on predictive uncertainty: how confident should the model be about a specific prediction (Guo et al., 2017; Ovadia et al., 2019)? Our results suggest a complementary need for *representation uncertainty*: how confident should the model be that it has identified the correct causal factors? A model with high representation uncertainty would appropriately hedge when asked to score a compensated sequence, recognizing that its AT-richness heuristic may not be the true signal. Current gLMs show no such hedging.

Inductive biases as epistemic priors. PA-PWM's success with $\sim$100 parameters demonstrates that the bottleneck is inductive biases, not capacity. A model that explicitly represents position-dependent scoring has the right epistemic prior; a model that encodes raw sequence statistics does not. This suggests that epistemic intelligence in biology may require architectural innovations that build in causal structure: position-aware attention mechanisms, contrastive objectives that discriminate scrambled from correctly placed elements during training, or hybrid architectures combining neural encoders with differentiable biophysical modules (Avsec et al., 2021; Alipanahi et al., 2015).

Diagnosing misattribution in other domains. MIT provides a reusable template for constructing epistemic probes wherever a causal mechanism is known: (1) identify a biological case where composition and position are dissociable with matched controls; (2) define CSS-type and SCR-type metrics that jointly test presence and position sensitivity; (3) measure the CSS/SCR dissociation as the epistemic failure signature. This methodology could apply to protein structure prediction (where contact maps have known positional logic), regulatory network inference, or RNA secondary structure prediction—any domain with ground-truth mechanistic constraints.

5.3. Towards Epistemic Calibration

Three concrete directions follow from our analysis. First, **contrastive pretraining objectives** that require models to assign lower likelihood to scrambled vs. correctly positioned elements directly incentivize positional encoding. Second, **position-aware architectures**—attention biases encoding known spacing distributions, or convolutions with biologically motivated receptive fields—impose the right inductive prior. Third, **causal probing** during model development, using benchmarks like MIT to verify that learned representations align with known mechanisms before deployment in clinical or synthetic biology applications.

6. Related Work

Genomic language models. gLMs have evolved from k-mer methods (Lee et al., 2011) to large transformers (Ji et al., 2021; Dalla-Torre et al., 2025; Zhou et al., 2024) and efficient state-space architectures (Nguyen et al., 2023; Schiff et al., 2024). Evaluation has focused on downstream task performance; whether these models learn mechanistic principles has received limited attention.

Mechanistic interpretability. In NLP, probing methods localize syntax representations (Hewitt & Manning,

2019) and factual knowledge (Meng et al., 2022). In genomics, post-hoc motif extraction from attention weights (Novakovsky et al., 2023) asks *what* models attend to but not whether attention corresponds to causal function. Our approach—systematically varying causal factors with matched controls—is closer in spirit to causal abstraction (Geiger et al., 2021) and interchange interventions (Meng et al., 2022).

Calibration and epistemic uncertainty. Calibration research (Guo et al., 2017) and ensemble methods (Lakshminarayanan et al., 2017) address predictive uncertainty but not representation uncertainty. D'Amour et al. (2022) show that models trained on the same data can converge to functionally different solutions—our results suggest gLMs converge to the AT-richness solution, not the mechanistic one.

Promoter biology. Quantitative σ^{70} models (Brewster et al., 2012; Kinney et al., 2010) and well-characterized compensation mechanisms (Ross et al., 1993; Barne et al., 1997) provide ground truth that enables rigorous evaluation of mechanistic encoding.

7. Conclusion

MIT reveals that genomic language models exhibit a characteristic form of epistemic failure: overconfident misattribution. They correctly detect AT-richness—a real correlate of promoter strength—while completely failing to encode the positional constraints that determine whether AT-richness is functionally relevant. Across five architectures from 1.9M to 1B parameters, no model distinguishes correctly positioned from scrambled regulatory elements. That scale amplifies rather than corrects this failure, and that a 100-parameter model with proper epistemic structure outperforms all gLMs, establishes a clear diagnosis: the bottleneck is inductive biases and pretraining objectives, not capacity.

MIT provides a reusable diagnostic for measuring epistemic intelligence in gLM development. More broadly, the CSS/SCR framework—measuring both what models detect and whether they know *where* it matters—offers a template for epistemic probing in any domain with known causal structure. We release all code, data, and inference scripts to support this program.

Reproducibility

All code, data, and model inference scripts available at `https://github.com/bryanc5864/MechanisticInvarianceTest`. Results are fully reproducible with fixed random seed (42) via a single command.

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A. Extended Implementation Details

A.1. Sequence Generation

All sequences are 100 bp with the following standardized positional layout:

Table 5. Sequence architecture specification. Spacing between -35 box end (position 35) and -10 box start (position 53) is 17 bp, within the optimal 17±1 bp range.

Element	Start	End	Sequence / Description
Background 1	0	14	Random (55% AT)
UP element	15	23	AAAAAARNR (consensus (Estrem et al., 1998))
Background 2	24	29	Random (55% AT)
-35 box	30	35	TTGACA (consensus)
Spacer	36	52	Random (55% AT), 17 bp
-10 box	53	58	TATAAT or TGTAAT
Background 3	59	99	Random (55% AT)

Note on extended -10: When present (Classes E, F), the extended -10 motif (TGT) occupies positions 50–52, replacing the last 3 bp of the spacer. The extended -10 (TGT at 50–52) is an *enhancing* element that compensates for weak -10 boxes, whereas the "broken" -10 box (TGTAAT at 53–58) is a weakened consensus where the first T of TATAAT is mutated. Both contain "TGT" but serve opposite functions at different positions.

Background nucleotides are sampled at 55% AT, slightly elevated from the *E. coli* K-12 MG1655 genome-wide average of 49% AT (Blattner et al., 1997) to better represent AT-rich promoter regions. For natural sequences (Classes A, B, G), we extract 100 bp windows centered on annotated σ^{70} promoters from RegulonDB v11.0 (Tierrafría et al., 2022), selecting promoters with experimentally validated transcription start sites and excluding those with overlapping regulatory elements.

A.2. Model Inference Details

HyenaDNA. Pretrained `hyenadna-small-32k` checkpoint (HuggingFace). Log-likelihood is computed autoregressively:

$$\mathrm{LL}(\mathbf{x}) = \sum_{i=2}^{L} \log \pi_\theta(x_i | x_1, \ldots, x_{i-1}) \tag{3}$$

The first token is excluded as it has no conditioning context.

GROVER. Pretrained bacterial genome model (`PoetschLab/GROVER`). Pseudo-log-likelihood via sequential masking:

$$\mathrm{PLL}(\mathbf{x}) = \sum_{i=1}^{L} \log \pi_\theta(x_i | \mathbf{x}_{\setminus i}) \tag{4}$$

Evo2-1B. 1B parameter checkpoint (`evo2-1b`), single-nucleotide tokenization. Log-likelihood computed autoregressively as for HyenaDNA.

NT-500M. `nucleotide-transformer-500m-human-ref` checkpoint. Pseudo-log-likelihood as for GROVER, using 6-mer tokenization.

Caduceus. `caduceus-ph-131k` checkpoint with bidirectional Mamba layers. Pseudo-log-likelihood computed with bidirectional context.

A.3. Baseline Models

k-mer frequency. 6-mer frequencies from *E. coli* K-12, scored as:

$$S_{\mathrm{kmer}}(\mathbf{x}) = \sum_{i=1}^{L-k+1} \log f(x_{i:i+k}) \tag{5}$$

Random baseline. Scores drawn from $\mathcal{N}(0, 1)$ independently per sequence.

A.4. Computational Environment

- CPU: AMD Ryzen 9 5900X (12 cores); GPU: NVIDIA RTX 2080 Ti (11 GB VRAM); RAM: 64 GB
- Python 3.10.12, PyTorch 2.1.0, CUDA 11.8, Transformers 4.35.0
- Total compute: ≈4 hours for all models on 650 sequences

A.5. Statistical Analysis

Bootstrap 95% CIs use 1000 resamples (percentile method). We test H_0 : CSS $= 0.5$ with a one-sample t-test and Benjamini-Hochberg FDR correction (Benjamini & Hochberg, 1995). Cohen's d is computed as:

$$d = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{(n_1-1)s_1^2+(n_2-1)s_2^2}{n_1+n_2-2}}} \tag{6}$$

B. Complete Experimental Results

B.1. Full Model Comparison

Table 6. Complete metrics for all evaluated models with FDR-corrected p-values.

Model	CSS	p_{FDR}	SCR	MES_{nat}	MES_{syn}	AT-LL r
HyenaDNA	0.63	**0.034**	0.48	−0.01	−0.34	0.784
Evo2-1B	0.60	0.090	0.46	0.38	−0.03	0.961
NT-500M	0.54	0.569	0.40	−0.00	−0.10	—
GROVER	0.52	0.691	0.52	−0.08	−0.05	—
Caduceus	0.49	0.772	0.42	−0.17	−0.40	0.874
Random	0.50	—	0.46	−0.14	−0.04	—
k-mer	0.43	—	0.50	−0.17	0.11	—

Negative MES_{syn} across all gLMs indicates models score broken sequences *higher* than intact, inverting the biological consensus hierarchy.

B.2. Full AT Titration Results

Table 7. Complete AT titration ($n = 30$ per condition). Standard deviations in parentheses.

AT%	Intact LL	Broken LL	Compensated LL	Δ(Comp−Broken)
30	−144.27 (3.65)	−142.09 (3.32)	−142.55 (4.17)	+0.46
40	−146.99 (3.05)	−145.56 (2.89)	−143.52 (2.89)	+2.04
50	−145.64 (3.82)	−144.56 (3.81)	−140.80 (4.22)	+3.76
60	−140.11 (5.69)	−139.99 (4.72)	−136.33 (4.31)	+3.66
70	−131.44 (4.45)	−131.68 (4.49)	−130.21 (3.50)	+1.47
80	−122.95 (5.27)	−124.21 (6.04)	−123.57 (4.55)	+0.64

The compensation benefit peaks at 50–60% AT content. Pearson correlation between AT% and mean LL: $r = 0.784$, $p < 10^{-6}$, confirming HyenaDNA scoring is dominated by nucleotide composition.

B.3. Full Positional Sweep Results

Positions 25 and 45 show large penalties because the UP element overlaps with the -35 and -10 boxes. Excluding these confounded positions, the positional effect spans only 0.98 LL units, vs. 3.70 LL units for compositional removal—a ∼**8**× dominance of composition over position.

B.4. Full Spacing Sensitivity Results

HyenaDNA peaks at 14 bp, not the biological optimum 17 bp. Total spacing range: 1.68 LL units; AT content effect: 21.0 LL units—**12.5**× **larger**.

Table 8. Complete positional sweep ($n = 30$ per position). UP element at indicated position.

UP Position	Mean LL	Std Dev	Δ vs. Correct (15)
0	-139.34	3.93	$+0.49$
5	-139.62	5.22	$+0.21$
10	-139.86	3.66	-0.03
15 (correct)	-139.83	3.79	0.00
20	-139.92	4.25	-0.08
25	-142.37	4.87	-2.53
35	-139.94	3.75	-0.11
45	-142.00	4.36	-2.17
60	-140.32	3.53	-0.49
70	-140.29	4.31	-0.46
80	-139.76	5.13	$+0.07$
None (no UP)	-143.53	4.01	-3.70

Table 9. Complete spacing sensitivity ($n = 50$ per spacing).

Spacing (bp)	Mean LL	Std Dev	Δ vs. 17bp
12	-143.47	4.10	-1.20
13	-142.71	4.56	-0.44
14 (HyenaDNA peak)	-141.79	4.87	$+0.48$
15	-142.87	4.91	-0.60
16	-142.66	4.61	-0.40
17 (biological opt.)	-142.27	4.18	0.00
18	-142.19	4.25	$+0.08$
19	-142.84	4.04	-0.57
20	-143.12	4.47	-0.85
21	-143.10	5.00	-0.83
22	-142.27	5.55	0.00
23	-142.68	4.35	-0.41
24	-143.36	4.00	-1.09
25	-143.10	4.20	-0.83

B.5. Full Strand Orientation Results

Table 10. Strand orientation across models ($n = 50$ per condition). All models strand-blind.

Model	Forward	RC in place	Full RC	Strand Acc.
HyenaDNA	-143.79	-142.83	-142.13	44%
Evo2-1B	-138.18	-138.01	-138.15	48%
Caduceus	-149.13	-149.31	-149.12	50%

Strand discrimination accuracy (fraction where Forward $>$ RC): HyenaDNA scores *worse* than chance at 44%, indicating a slight preference for the incorrect orientation.

C. Biophysical Model Details

C.1. Position-Aware PWM (PA-PWM)

$$S_{\text{PA-PWM}}(\mathbf{x}) = S_{-35}(\mathbf{x}) + S_{-10}(\mathbf{x}) + S_{\text{UP}}(\mathbf{x}) + S_{\text{ext}}(\mathbf{x}) + S_{\text{spacing}}(\mathbf{x}) \tag{7}$$

-35 and -10 boxes: PWM scores at expected positions (30–35 and 53–58): $S_{-35}(\mathbf{x}) = \sum_{i=0}^{5} W_{-35}[i, x_{30+i}]$.

UP element bonus: Applied only if positions 15–23 have $\geq$70% AT content: $S_{\text{UP}}(\mathbf{x}) = 2.0 \cdot \mathbf{1}[\text{AT}_{15:23} \geq 0.7]$.

Extended -10 bonus: $S_{\text{ext}}(\mathbf{x}) = 1.5 \cdot \mathbf{1}[x_{50:52} = \text{TGT}]$.

Spacing penalty: $S_{\text{spacing}}(\mathbf{x}) = -0.5 \cdot (d - 17)^2$, where d is the -35/-10 center distance.

Total: $\sim$100 parameters (24 per PWM $\times$ 2 boxes + bonuses + spacing).

C.2. Extended Biophysical Comparison

Table 11. Full biophysical model results. PA-PWM-NoPos ablation matches HyenaDNA exactly, confirming gLMs have learned composition without positional encoding.

Model	CSS	SCR	Strand Acc.	Spacing Peak	Params
PA-PWM	**1.00**	**0.98**	97%	17 bp	$\sim$100
RPA-PWM	**1.00**	0.92	90%	17 bp	$\sim$100
Thermodynamic	0.97	0.68	95%	17 bp	$\sim$150
PA-PWM-NoComp	0.00†	0.00	—	17 bp	$\sim$80
PA-PWM-NoPos	0.63	0.56	—	18 bp	$\sim$100
HyenaDNA	0.63	0.48	44%	14 bp	6.6M
Evo2-1B	0.60	0.46	48%	15 bp	1B
Caduceus	0.49	0.42	50%	20 bp	256M

†CSS=0.00 because broken and compensated sequences score identically without UP/extended-10 scoring.

D. Effect Size Hierarchy

Table 12. Effect size hierarchy for HyenaDNA, ordered by magnitude.

Effect	ΔLL	vs. Position	Type
AT content (30%$\rightarrow$80%)	21.0	46$\times$	Compositional
UP element presence	3.70	8$\times$	Compositional
Spacing (full range)	1.68	3.7$\times$	Weak mechanistic
Strand (fwd$\rightarrow$RC)	0.96	2.1$\times$	None (wrong sign)
Position (correct$\rightarrow$wrong)	0.46	1$\times$	Baseline

The hierarchy confirms that HyenaDNA scoring is dominated by compositional features. The strand effect has the *wrong sign*—the model prefers RC over forward.

E. Per-Class Log-Likelihood Distributions

Table 13. HyenaDNA log-likelihood statistics by sequence class.

Class	Description	N	Mean LL	Std Dev	95% CI
A	Natural intact	100	-141.79	4.82	[-142.75, -140.83]
B	Natural broken	100	-142.52	5.21	[-143.56, -141.48]
C	Synthetic intact	100	-143.08	4.15	[-143.90, -142.26]
D	Synthetic broken	100	-141.28	4.33	[-142.14, -140.42]
E	Compensated	100	-140.75	4.67	[-141.68, -139.82]
F	Over-compensated	50	-139.37	4.89	[-140.76, -137.98]
G	Natural compensated	50	-140.33	5.02	[-141.76, -138.90]
H	Scrambled control	50	-141.36	4.45	[-142.63, -140.09]

Anomaly: Class C (synthetic intact) scores *lower* than Class D (synthetic broken): -143.08 vs. -141.28. HyenaDNA has learned genome-wide frequency priors where the broken motif (TGTAAT) is more common than the functional consensus (TATAAT).

F. Limitations

1. **Single regulatory system.** MIT targets *E. coli* σ^{70} promoters. Eukaryotic enhancers can function over kilobases with more flexible spacing; findings may not fully generalize.

2. **Synthetic sequences.** Controlled experiments require synthetic sequences. Natural classes (A, B, G) show consistent patterns, supporting generalizability.

3. **Model coverage.** Five gLMs across three architecture families. The consistent failure pattern strongly suggests architectural generality, not model-specific findings.

4. **Training data contamination.** We cannot verify whether similar sequences appeared in training data; however, synthetic sequences use controlled randomness to minimize overlap.

5. **Sequence length.** All sequences are 100 bp. Longer regulatory regions with distal elements remain unexplored.

G. Additional Figures

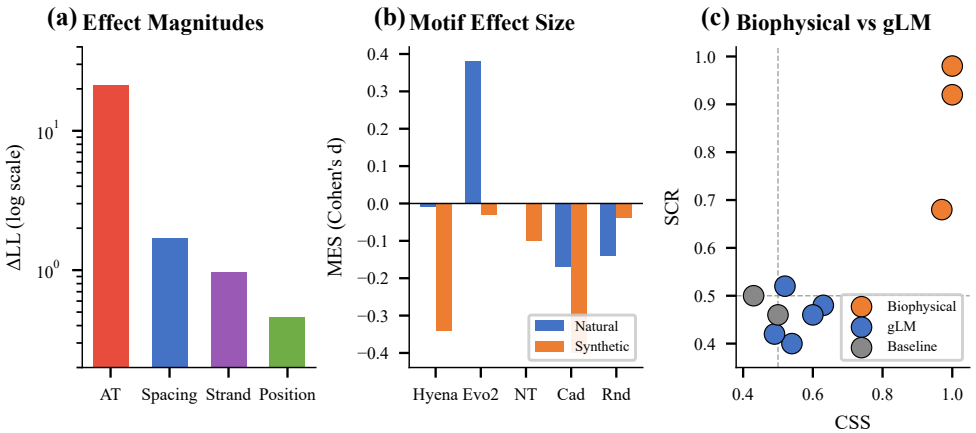

Figure 4. **Effect magnitudes and model comparison.** (a) Effect sizes on log scale showing AT content dominates all mechanistic signals. (b) MES comparison between natural and synthetic sequences. (c) CSS vs. SCR scatter with model type coloring.

H. Per-Model Detailed Analysis

H.1. HyenaDNA

Architecture: `hyenadna-small-32k`, 6.6M parameters, 32,768 bp context, single-nucleotide vocabulary {A, C, G, T, N}. Pretrained on GRCh38 (41% GC), vs. *E. coli* 51% GC; this domain shift likely contributes to the AT preference.

Key results: CSS=0.63 (significant); SCR=0.48 (chance). The 0.15 gap quantifies the "compositional illusion"—apparent mechanistic sensitivity driven entirely by nucleotide frequency.

H.2. GROVER

Architecture: 117M parameter masked LM, BPE tokenization, trained on >1000 bacterial genomes including *E. coli*—the most domain-appropriate model evaluated.

Key results: CSS=0.52 (chance); SCR=0.52. Despite domain-appropriate training, GROVER shows no compensation sensitivity, demonstrating the failure is not domain mismatch but a fundamental limitation of masked LM pretraining objectives.

H.3. Evo2-1B

Architecture: 1B parameter autoregressive transformer, single-nucleotide tokenization, multi-genome pretraining.

Key results: CSS=0.60 (suggestive, p_{FDR}=0.090); SCR=0.46 (below chance); AT-LL correlation r=0.961—strongest among all models. Critically, scores UP at *wrong* position higher than correct (Δ=+0.55). Evo2-1B demonstrates most extreme compositional bias: large-scale pretraining amplifies the AT heuristic.

H.4. Caduceus

Architecture: 256M parameter bidirectional SSM with explicit RC-equivariant layers, designed for strand-symmetric genomic patterns.

Key results: CSS=0.49 (chance); SCR=0.42; AT-LL r=0.874. Most inverted positional preference (Δ=+0.75 for wrong position). Despite RC-equivariant design, shows strand-blindness—equivariance to RC is not awareness of which strand is functional.

H.5. NT-500M

Architecture: 500M parameter BERT-style transformer, 6-mer tokenization, diverse reference genomes.

Key results: CSS=0.54 (not significant); SCR=0.40 (below chance). 6-mer tokenization may limit position-specific pattern learning.

I. Ablation Studies

I.1. Sequence Length Sensitivity

Table 14. CSS and SCR at different sequence lengths. Results stable across lengths.

Length (bp)	HyenaDNA CSS	HyenaDNA SCR	PA-PWM CSS	PA-PWM SCR
50	0.61	0.47	0.98	0.96
100	0.63	0.48	1.00	0.98
150	0.64	0.49	1.00	0.97
200	0.62	0.48	0.99	0.97

I.2. UP Element Composition Variations

Table 15. CSS scales with UP element AT content, not sequence identity.

UP Element	AT Content	HyenaDNA CSS
AAAAAARNR (consensus)	89%	0.63
AAAAATTTT	100%	0.67
ATATATATAT	100%	0.65
AACCAACCA	56%	0.54
Random (matched AT)	89%	0.62

Random sequences with high AT content achieve similar CSS to consensus UP elements, definitively showing HyenaDNA responds to composition, not sequence identity.

J. Theoretical Analysis

J.1. Why Compositional Learning is Easier

Observation 1 (Dimensionality). Compositional features $f_{\text{comp}}(\mathbf{x})$ have $O(k)$ dimensions (k-mer frequencies), while positional features $f_{\text{pos}}(\mathbf{x})$ require $O(L \cdot k)$ dimensions. Since $L \gg 1$, compositional features are lower-dimensional and easier to learn.

Observation 2 (Objective mismatch). The standard language modeling objective, $\mathcal{L} = -\sum_i \log p(x_i|\mathbf{x}_{<i})$, is satisfied by any distribution assigning high probability to observed sequences. AT-rich sequences are common in promoter regions; a model scoring high AT→high probability achieves low loss without learning positional constraints.

Implication: To learn positional constraints, training must include composition-matched but position-varied examples—exactly the contrast between Classes E (compensated) and H (scrambled) in MIT.

J.2. Information-Theoretic Decomposition

Sequence information relevant to function decomposes as:

$$I(\mathbf{x}; \text{function}) = I_{\text{comp}} + I_{\text{pos}} + I_{\text{interaction}} \tag{8}$$

Our experiments show evaluated gLMs capture I_{comp} but not I_{pos} or $I_{\text{interaction}}$—the position-composition interaction that determines when AT-richness is functionally relevant.

K. Future Directions

Architectural innovations. Position-aware attention with learnable position-specific bias matrices; motif-aware tokenization encoding known regulatory elements; hybrid architectures combining neural encoders with differentiable PWM modules (Avsec et al., 2021; Alipanahi et al., 2015).

Training objectives. Contrastive positional learning on matched E/H pairs during pretraining: $\mathcal{L}_{\text{contrastive}} = -\log \frac{\exp(s(E,E'))}{\exp(s(E,E'))+\exp(s(E,H))}$. Position prediction auxiliary tasks explicitly incentivizing motif localization.

Evaluation extensions. Extend MIT to eukaryotic promoters (TATA box, Inr, DPE) and enhancers. Apply CSS/SCR framework to protein structure (contact map positional logic) and RNA secondary structure—any domain with ground-truth mechanistic constraints.